Fiscal Year:	FY 2016	**Task Last Updated:**	FY 10/13/2016
PI Name:	Williams, Forman A. Ph.D.		
Project Title:	Flammability Limit Experiment (FLEX)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:			
City:	La Jolla	**State:**	CA
Zip Code:	92093-0411	**Congressional District:**	53
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	08/01/2007	**End Date:**	07/31/2016
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Hicks, Michael C.	**Contact Phone:**	216-433-6576
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dryer, Frederick Ph.D. (Experiment Co-I: Princeton University) Choi, Mun Young Ph.D. (Experiment Co-I: University of Connecticut) Shaw, Benjamin Ph.D. (Experiment Co-I: University of California, Davis) Avedisian, C. Thomas Ph.D. (Experiment Co-I: Cornell University) Farouk, Tanvir Ph.D. (Experiment Co-I: University of South Carolina) Hicks, Michael M.S. (Experiment Co-I: NASA Glenn Research Center) Dietrich, Daniel Ph.D. (Experiment Co-I: NASA Glenn Research Center) Nayagam, Vedha Ph.D. (Experiment Co-I: Case Western Reserve University)		
Grant/Contract No.:	NNX09AM33A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

The overall objective of the FLEX experiments (Flammability Limit Experiment) is to improve fundamental knowledge of droplet burning and other combustion processes relevant to fire safety in manned space environments. Specific objectives are to investigate influences of low-gravity environments on flammability conditions of different fuels in different oxidizing atmospheres representative of fire-suppression scenarios.
The first phase of FLEX was a focused set of science objectives designed to address spacecraft fire safety concerns. The primary objectives were the assessment of flammability limits in a range of habitable atmospheres and fire suppressant efficacies. The second phase is FLEX-2 with a more expansive set of science objectives designed to investigate fundamental phenomena associated with (i) fuel sooting, (ii) development of reference fuels for fuel surrogates, (iii) low flow convective effects on diffusion flames, (iv) spacecraft fire safety, (v) droplet interactions in spray combustion, and (vi) condensed phase effects on burning using bi-component fuel blends.

The research team for FLEX comprises externally funded Principal Investigators (PIs) and internally funded PIs. The lead researcher is Forman Williams for both FLEX and FLEX-2 and can be considered the de facto “PI” for FLEX (i.e., the combined experiments). The externally funded PIs are grantees affiliated with universities. The internally funded PIs are Michael C. Hicks (NASA-Glenn Research Center-GRC), Daniel L. Dietrich (NASA-GRC), and Vedha Nayagam (Case Western Reserve University).

Flammability Limit Experiment (FLEX) Experiment PI: Forman Williams, Ph.D., University of California San Diego.

FLEX and FLEX-2 Experiment External CoInvestigators (CoIs):

Frederick Dryer, Ph.D., Princeton University. Task title: "Scientific Support for a Space Based Droplet Burning Experiment"

Mun Young Choi, Ph.D., University of Connecticut. Task title: "FLEX Droplet Combustion Experiments in Microgravity"

FLEX-2 Experiment External CoInvestigators (CoIs):

C. Thomas Avedisian, Ph.D., Cornell University. Task title: "Combustion of Surrogate Fuel Droplets in Microgravity"

Benjamin Shaw, Ph.D., University of California Davis. Task title: "Scientific Support for the FLEX-2 ISS (International Space Station) Experiment"

Tanvir Farouk, Ph.D., University of South Carolina. Task title: "Mathematical Modeling and Simulation of Space ased Droplet Combustion – Scientific Support to FLEX and FLEX-2 Program"

FLEX and FLEX-2 Experiment Internal CoInvestigators (CoIs):

Michael C. Hicks, MSME, NASA Glenn Research Center.

Daniel L. Dietrich, Ph.D., NASA Glenn Research Center.

Vedha Nayagam, Ph.D., Case Western Reserve University.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The overall objective of the FLEX experiments (Flammability Limit Experiment) was to improve fundamental knowledge of droplet burning and other combustion processes relevant to fire safety in manned space environments. Specific objectives were to investigate influences of low-gravity environments on flammability conditions of different fuels in different oxidizing atmospheres representative of fire-suppression scenarios. Experiments with hepane as the fuel and with methanol as the fuel were performed aboard the ISS.
The project, in general, involved basic flammability-limit investigations relevant to fire-safety issues for manned space flight. In standard fire-suppression techniques, a suppressant jet is directed towards a fire or a smoldering region. In considering these techniques for space applications, it is necessary to analyze the effect of suppressant transport to both the near-field and far-field of the region of interest. Fire prevention can also be implemented by modifying the atmospheric composition of the environment so as to reduce the likelihood of a self-sustaining combustion event. Finally, the flammability properties of materials can also be modified by the addition of materials which suppress ignitability, enhance charring of surfaces, or chemically inhibit self-sustained burning. The research addressed these aspects of fire safety in space.

Some general directions of thrusts were as follows:

1. Map the flammability boundaries for liquid fuel combustion in reduced gravity.

2. Quantify the suppressant effectiveness of various gaseous suppressants over the range of candidate atmospheric pressures and O2 concentrations.

3. Develop predictive theoretical analyses and chemical-kinetic schemes to model flammability boundaries as a function effective gravitational acceleration on the unique ambient conditions encountered in space-exploration applications.

4. Devise improved and validated reduced (simplified) sub-models of important physical processes (chemical kinetics, etc.) that can be used in simulations of large scale, ‘realistic’ fires.

5. Determine combustion characteristics of liquid-fuel droplets of ethanol, decane, and liquid-fuel mixtures in varying atmospheres that can improve understanding of means for achieving clean and efficient power production.

6. Analyze cool-flame quasi-steady combustion of alkane droplets, especially heptane, but also decane and octane. The research at University of California, San Diego (UCSD) resulted in a number of publications, including:

1. J. Dzik, V. Nayagam, and F. A. Williams, “Ignition and Combustion of n-Heptane Droplets in Carbon Dioxide Enriched Environments,” International Communications in Heat and Mass Transfer 37, 221-225 (2010).

2. M. C. Hicks, V. Nayagam, and F. A. Williams, “Methanol Droplet Extinction in Carbon-Dioxide-Enriched

Task Progress:	Environments in Microgravity," Combustion and Flame 157, 1439-1445 (2010). 3. J. Urzay, V. Nayagam and F. A. Williams, "Theory of the Propagation Dynamics of Spiral Edges of Diffusion Flames in von Karman Swirling Flows," Combustion and Flame 158 255-272 (2011). 4. V. Nayagam and F. A. Williams, "Analysis of the Melt Phase of a Rotating Polymer Disc Supporting a Diffusion Flame," Journal of Fluid Mechanics 687, 238-253 (2011). 5. V. Nayagam, D. L. Dietrich, P. V. Ferkul, M. C. Hicks, and F. A. Williams, "Can Cool Flames Support Quasi-Steady Alkane Droplet Burning?", Combustion and Flame 159, 3583-3588 (2012). 6. G. Paczko, N. Peters, K. Seshadri, and F.A. Williams, "The role of cool-flame chemistry in quasi-steady combustion and extinction of n-heptane droplets", Combustion Theory and Modelling 18, 515–531(2014). 7. D. L. Dietrich, V. Nayagam, M. C. Hicks, P. V. Ferkul, F. L. Dryer, T. Farouk, B. D. Shaw, H. K. Su, M. Y. Choi, Y. C. Liu, C. T. Avedisian, and F. A. Williams "Droplet Combustion Experiments Aboard the International Space Station", Microgravity Sicence and Technology 26, 65–76 (2014). 8. V. Nayagam, D. L. Dietrich, M. C. Hicks, and F. A. Williams, "Cool-Flame Extinction During n-Alkane Droplet Combustion in Microgravity," Combustion and Flame 162, 2140- 2147 (2015). 9. V. Nayagam, D. L. Diertich, and F. A. Williams, "Partial-Burning Regime for Quasi-Steady Droplet Combustion Supported by Cool Flames", AIAA Journal 54, 1235-1239, (2016). 10. K. Seshadri, N. Peters, F. A. Williams, V. Nayagam, and G. Paczko, "Asymptotic Analysis of Quasi-Steady n-Heptane Droplet Combustion Supported by Cool-Flame Chemistry", Combustion Theory and Modelling, to appear, 2016. In addition, preparation of other manuscripts is in progress. The first two publications report analyses of experiments performed in ISS, while the next two report analyses of ground-based experiments, in all cases with the relationships to fire safety kept in mind. The fifth publication is the formal announcement of the discovery of quasi-steady droplet combustion supported solely by cool-flame chemistry in the ISS experiments, which was not expected in advance and which made news and resulted in the planned CFI (Cool Flame Investigation) experiments. The next entry is a computational study of how this phenomenon can occur, and the last citation reports corresponding analytical developments describing this phenomenon. The penultimate work listed is a theoretical investigation preparing for future analyses of the structures of these cool flames, while the one before that reports further experimental results concerning these cool flames, from measurements made in ISS, along with their interpretation. Finally, the seventh paper is a report from the entire FLEX team concerning the full range of results flowing from the program. Citations to these publications and others from the project are listed in the Cumulative Bibliography. NOTE: Project continues as "Flammability Limit Experiment (FLEX) - Phase II," grant NNX16AR09A. See that project for subsequent reporting.

Bibliography Type:	Description: (Last Updated: 03/07/2024)
Articles in Peer-reviewed Journals	Hicks MC, Nayagam V, Williams FA. "Methanol droplet extinction in carbon-dioxide-enriched environments in microgravity." Combustion and Flame. 2010 Aug;157(8):1439-45. http://dx.doi.org/10.1016/j.combustflame.2010.05.007 , Aug-2010
Articles in Peer-reviewed Journals	Urzay J, Nayagam V, Williams FA. "Theory of the propagation dynamics of spiral edges of diffusion flames in von Kármán swirling flows." Combustion and Flame. 2011 Feb;158(2):255-72. http://dx.doi.org/10.1016/j.combustflame.2010.08.015 , Feb-2011
Articles in Peer-reviewed Journals	Nayagam V, Williams FA. "Analysis of the melt phase of a rotating polymer disc supporting a diffusion flame." Journal of Fluid Mechanics. 2011 Nov;687:238-53. http://dx.doi.org/10.1017/jfm.2011.349 , Nov-2011
Articles in Peer-reviewed Journals	Nayagam V, Dietrich DL, Williams FA. "Partial-burning regime for quasi-steady droplet combustion supported by cool flames." AIAA Journal. 2016 Apr;54(4)1235-9. http://dx.doi.org/10.2514/1.J054437 , Apr-2016
Articles in Peer-reviewed Journals	Seshadri K, Peters N, Williams FA, Nayagam V, Paczko G. "Asymptotic analysis of quasi-steady n-heptane droplet combustion supported by cool-flame chemistry." Combustion Theory and Modelling. 2016;20(6):1118-30. http://dx.doi.org/10.1080/13647830.2016.1248494 , Nov-2016
Articles in Peer-reviewed Journals	Shaw BD, Vang CL. "Oxygen lewis number effects on reduced gravity combustion of methanol and n-heptane droplets." Combustion Science and Technology. 2016 Jan 2;188(1):1-20. https://doi.org/10.1080/00102202.2015.1072176 , Jan-2016